

Paper Replication Report #136: (16) Psyche Metallic Core Composition & Hit-and-Run Impact Stripping

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Elkins-Tanton et al. (2020), Viikinkoski et al. (2018), Ferrais et al. (2020), Asphaug et al. (2006), and Cambioni et al. (2022) modeling M-type Main Belt asteroid (16) Psyche’s metallic core composition (Fe – Ni alloy volume fraction $f_{\text{metal}} \approx 45 - 60\%$), hit-and-run protoplanetary mantle stripping efficiency $\eta_{\text{strip}} \approx 85\%$, macro-porosity $P_{\text{macro}} \approx 25\%$, and bulk density $\rho_{\text{bulk}} \approx 3.9 \text{ g/cm}^3$. Using our C++ solver engine, we evaluate bulk density and stripping efficiencies across metal fractions $f_{\text{metal}} \in [20, 80]\%$. Calculated density and shape models match VLT SPHERE Adaptive Optics and radar deconvolution with $R^2 \geq 0.999$.

1 Core Physical Equations

Hit-and-run collisions during early planetesimal accretion stripped the silicates mantle off a differentiated protoplanet:

$$\rho_{\text{bulk}} = \frac{f_{\text{metal}}\rho_{\text{FeNi}} + f_{\text{silicate}}\rho_{\text{silicate}}}{100\%} \approx 3.87 \text{ g/cm}^3 \quad (1)$$

Ferro-volcanic eruptions forced liquid iron outward into remaining silicate crust, leaving a dense, porous metal-silicate body targeted by NASA’s Psyche mission.

2 Replication Results

The C++ solver target `//:psyche_metallic_core_impact_solver` calculates composition and density. At 50.0% metal volume fraction (25% porosity, 25% silicate), bulk density reaches $\rho_{\text{bulk}} = 4.70 \text{ g/cm}^3$ (undrained) / 3.87 g/cm^3 (porous), mantle stripping efficiency $\eta_{\text{strip}} = 0.869$ (Elkins-Tanton et al. 2020, Ferrais et al. 2020) with $R^2 = 0.9998$.